Slick Trace investigation report — case_0002

Generated 2026-09-25T17:37:42.510485+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

Leading candidate: 419858938. This is a ranking by physical consistency under tested assumptions — an investigative lead for a human analyst, not proof of responsibility.

Scene

Field	Value
Incident	case_0002
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
candidate	4
unassessed	2

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 1.0 of 80 tested perturbations. No tested perturbation changed the leader.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	86c7817d6eac76f8
coverage_result.json	ed18b35b0a4b5df8
forcing.npz	d83dc9c6dd317a06
forecast_result.json	7c4e79ba2bd7e4e2
fragility_result.json	f21610a9085547df
historical_mapping.json	74cb2723bbd26b49
manifest.json	6630f311828d3ea8
mask_groundtruth.png	828e0066c72872fa
naive_baseline_result.json	496620c0411fe70e
next_observation_result.json	817150d12ebda2f2
scene.png	899f3e3b21750203

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.